

## LESSON 23 / 第二十三課

# Urban Tree Mortality: From Planting Programmes to Stewardship Covariates

## 城市樹木死亡：從種植計劃到管護共變項

*Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.*

*體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。*

### Abstract

#### 摘要

Urban canopy policy is often written as a planting target. The demographic literature shows that the binding constraint is survival. Hilbert et al. reviewed 56 studies: planting-cohort annual mortality 0.6–68.5% (medians 4.4–6.5%); repeated mixed-age inventories 0–30% (medians 2.3–2.6%). Mortality is front-loaded: first-five-year medians 6.6–7.0%, then 2.8–3.8%. Roman and Scatena estimated annual survival 94.9–96.5%, half-life 13–20 years and mean life expectancy 19–28 years — longer than the widely repeated 7- or 13-year claim. Demographic reviews rank taxon, size/age, site and stewardship. Tubby found that 81% of published dieback case studies named a biotic primary cause while discussing urban stress as predisposition. The two literatures sample different questions. Planting numbers without an establishment survival budget overstate future canopy.

城市樹冠政策常寫成種植目標。人口學文獻顯示，真正的約束是存活。Hilbert 等綜述 56 項研究：種植隊列年死亡率 0.6–68.5%（中位數 4.4–6.5%），不均齡重複清查 0–30%（中位數 2.3–2.6%）。死亡前置：首五年中位數 6.6–7.0%，其後 2.8–3.8%。Roman 與 Scatena 估計年存活率 94.9–96.5%、半衰期 13–20 年、平均預期壽命 19–28 年——長於廣為傳誦的 7 年或 13 年主張。人口學綜述排序分類群、大小／年齡、立地與管護。Tubby 發現 81% 已發表枯死案例以生物性為主因，同時把城市逆境討論為誘因。兩支文獻抽樣不同問題。沒有定植存活預算的種植數字會高估未來樹冠。

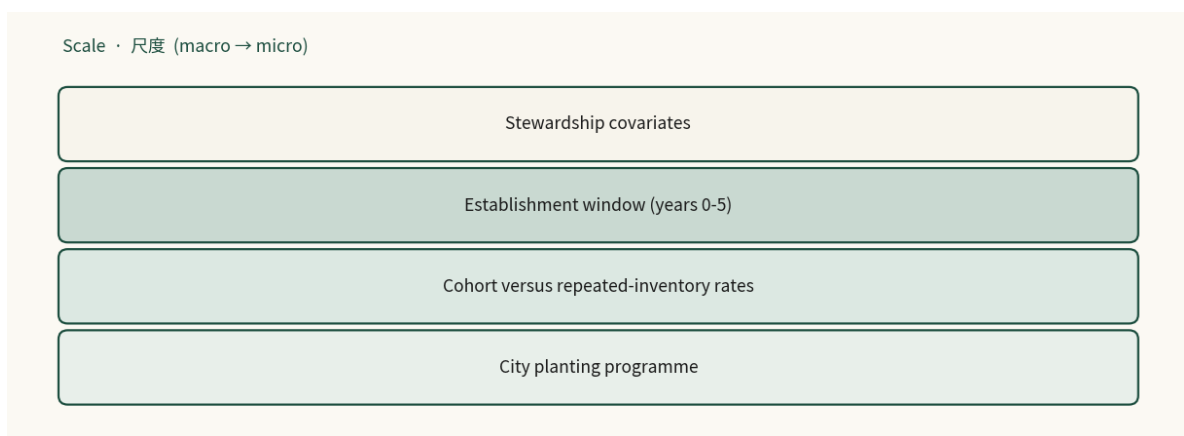


Figure 1. Nested scales from the planting programme to stewardship covariates.

圖 1. 由種植計劃到管護共變項的嵌套尺度。

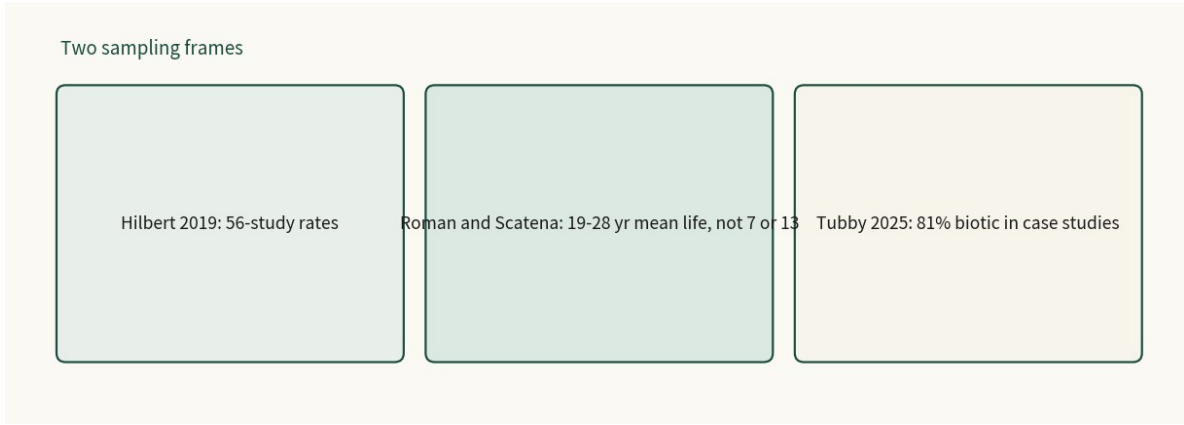


Figure 2. Three published results that a single folklore life-span hides.

圖2. 單一傳聞壽命所藏起的三項已發表結果。

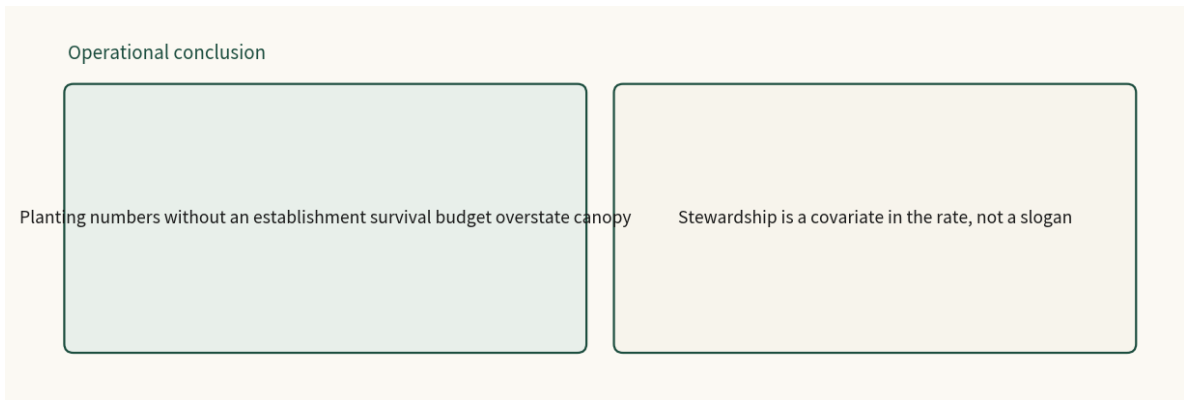


Figure 3. A planting target is not a canopy forecast.

圖3. 種植目標不是樹冠預測。

## 1. Scope

### 1. 範圍

The object is published urban-tree demography: rates, sampling frames and covariates. Folklore life-spans of 7 or 13 years are treated as claims to be tested, not as inputs.

對象是已發表的城市樹木人口學：率、抽樣框與共變項。7 年或 13 年的傳聞壽命被當成待檢驗主張，不是輸入。

## 2. Rates and the 7-/13-year claim

### 2. 率與 7／13 年主張

Hilbert et al. separated planting-cohort studies from repeated inventories of mixed-age populations and reported different median ranges for each [1]. The establishment window carries higher annual mortality than later years [1]. Roman and Scatena corrected the 7- and 13-year folklore: their meta-analysis implied mean life expectancies of 19–28 years and half-lives of 13–20 years [2]. Ko and colleagues' 22-year San Francisco cohort (42.4% dead) is one long window inside that literature, not a universal constant [3].

Hilbert 等把種植隊列研究與不均齡族群的重複清查分開，並為每一類報告不同中位數範圍 [1]。定植窗的年死亡率高於往後年份 [1]。Roman 與 Scatena 校正了 7 年與 13 年傳聞：他們的統合分析意味平均預期壽命 19–28 年、半衰期 13–20 年 [2]。Ko 及同事 22 年舊金山隊列（42.4% 死亡）是該文獻中的一個長窗，不是普遍常數 [3]。

### 3. Two literatures on cause

#### 3. 兩支關於成因的文獻

Demographic reviews most often rank taxon, size or age, site, and human stewardship or vandalism [1,3]. Tubby's systematic review of published dieback case studies found that 81% concluded an arthropod or microorganism was the primary cause, while frequently discussing urban stress as predisposition [4]. The two literatures sample different questions. Stewardship is a covariate in the rate [5,6], not a slogan pasted on after the rate is chosen.

人口學綜述最常排序分類群、大小或年齡、立地，以及人為管護或破壞 [1,3]。Tubby 對已發表枯死案例的系統綜述發現，81% 以節肢動物或微生物為主因，同時常把城市逆境討論為誘因 [4]。兩支文獻抽樣不同問題。管護是率中的共變項 [5,6]，不是在選好率之後貼上的口號。

### 4. Limits

#### 4. 限度

Most compiled studies are temperate North American and European streets. Tropical and arid-city rates are thinner. A median is not a specification for one street. Planting numbers without an establishment-phase survival budget overstate future canopy.

多數被彙編研究是溫帶北美與歐洲街道。熱帶與乾旱城市的率較薄。中位數不是一條街的規格。沒有定植期存活預算的種植數字會高估未來樹冠。

### Study questions

#### 研習題

*Choose the option that retains the published result. Answers follow the last item.*

選出保留已發表結果的選項。答案在最後一題之後。

#### 1. The binding constraint in this review is:

1. 本綜述中的約束是：

A. The planting announcement.

A. 種植宣布。

B. Survival.

B. 存活。

C. Flower colour.

C. 花色。

D. t/R.

D. t/R。

#### 2. Hilbert et al. treated cohort and inventory rates as:

2. Hilbert 等把隊列與清查率視為：

A. Interchangeable.

A. 可互換。

B. Different sampling frames with different medians.

B. 具不同中位數的不同抽樣框。

C. Both P50.

C. 都是 P50。

D. Both Wall 4.

D. 都是第 4 壁。

**3. First-five-year mortality relative to later years is:**

3. 相對往後年份，首五年死亡率是：

A. Lower.

A. 較低。

B. Elevated.

B. 升高。

C. Zero.

C. 為零。

D. Equal to flash density.

D. 等於閃擊密度。

**4. Roman and Scatena mean life expectancy was:**

4. Roman 與 Scatena 的平均預期壽命是：

A. 7 or 13 years.

A. 7 或 13 年。

B. 19-28 years.

B. 19–28 年。

C. 2 years.

C. 2 年。

D. 100 years exactly.

D. 正好 100 年。

**5. Tubby 2025 found biotic primaries in:**

5. Tubby 2025 發現生物性主因於：

A. 0% of case studies.

A. 0% 案例。

B. 81% of case studies.

B. 81% 案例。

C. All demographic reviews.

C. 所有人口學綜述。

D. Palms only.

D. 只有棕櫚。

**6. The two literatures:**

6. 兩支文獻：

A. Contradict each other on the same sample.

A. 在同一樣本上互相矛盾。

B. Sample different questions.

B. 抽樣不同問題。

C. Equal 3-30-300.

C. 等於 3-30-300。

D. Equal QTRA.

D. 等於 QTRA。

**7. Stewardship in this review is:**

7. 本綜述中的管護是：

- A. A slogan after the fact.
- A. 事後口號。
- B. A covariate in the rate.
- B. 率中的共變項。
- C. A lightning path.
- C. 雷電路徑。
- D. A cable grade.
- D. 鋼纜等級。

**8. A planting target without a survival budget:**

8. 沒有存活預算的種植目標：

- A. Forecasts canopy exactly.
- A. 精確預測樹冠。
- B. Overstates future canopy.
- B. 高估未來樹冠。
- C. Raises P88.
- C. 提高 P88。
- D. Is Wall 1.
- D. 是第 1 壁。

**9. Ko et al. 42.4% dead in 22 years is:**

9. Ko 等 22 年 42.4% 死亡是：

- A. A universal constant.
- A. 普遍常數。
- B. One long cohort inside the literature.
- B. 文獻中的一個長隊列。
- C. Proof of the 7-year claim.
- C. 7 年主張的證明。
- D. A tomogram.
- D. 斷層圖。

**10. A median urban mortality rate:**

10. 城市死亡率中位數：

- A. Is a specification for one street.
- A. 是一條街的規格。
- B. Is not a specification for one street.
- B. 不是一條街的規格。
- C. Equals occupancy.
- C. 等於佔用。
- D. Equals SOS.
- D. 等於 SOS。

**Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B**

## References

### 參考文獻

- [1] Hilbert D.R. et al. (2019). Urban tree mortality: a literature review. AUF 45: 167-200.
- [2] Roman L.A., Scatena F.N. (2011). Street tree survival rates. UFUG 10: 269-277.
- [3] Ko Y. et al. 22-year San Francisco cohort mortality.
- [4] Tubby K. et al. (2025). systematic review of urban dieback case studies.
- [5] Nowak D.J., McBride J.R., Beatty R.A. mortality and condition covariates.
- [6] Sklar F., Ames R.G. (1985). street-tree survival and participation.
- [7] Roman L.A. et al. young street-tree mortality in New York City.

*Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.*

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。